

ThinkFirst: An Adaptive Human AI Self-Regulation Framework for Promoting Responsible Generative AI Usage Through an Intelligent Browser Extension

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Abstract

Generative AI has rapidly transformed the way individuals perform academic, professional, and creative tasks by improving efficiency, accessibility, and productivity. However, the increasing reliance on AI-generated responses has raised concerns regarding the decline of independent thinking, critical reasoning, and cognitive engagement. Existing studies primarily focus on enhancing AI performance, while limited attention has been given to promoting responsible Human AI collaboration and reducing AI overdependence. This study proposes ThinkFirst, an intelligent browser extension designed to encourage responsible Generative AI usage through adaptive behavioral interventions. The proposed solution integrates the Adaptive Human AI Self-Regulation Framework (AHSRF), the AI Dependency Index (ADI), and the Adaptive Reflection Engine (ARE) to monitor user interaction behaviors, estimate AI dependency levels, and provide personalized reflection prompts before users rely on AI-generated responses. A preliminary survey was conducted to investigate current AI usage patterns and identify behavioral indicators associated with AI dependency, forming the basis for the proposed framework and system design. The proposed prototype demonstrates the feasibility of combining behavioral monitoring, privacy-preserving dependency assessment, and adaptive self-regulation within a browser extension environment without disrupting users' normal workflows. By emphasizing cognitive engagement rather than restricting AI usage, ThinkFirst contributes a practical human-centered framework for promoting responsible Human AI collaboration while maintaining the productivity benefits of Generative AI. The proposed approach is expected to support healthier AI usage habits and serve as a foundation for future intelligent behavioral intervention systems.

Keywords: *Generative AI, Responsible AI, Human AI Collaboration, Browser Extension, AI Dependency, Self-Regulation, Adaptive Intervention*

I. Introduction

Artificial Intelligence (AI), particularly Generative AI (GenAI), has fundamentally transformed how individuals acquire knowledge, solve problems, and perform knowledge-intensive tasks across education, research, industry, and everyday life. The rapid advancement of Large Language Models (LLMs), such as ChatGPT and other AI-powered assistants, enables users to generate content, retrieve information, support programming activities, and automate various cognitive processes with unprecedented speed and accessibility. Consequently, AI has evolved beyond a conventional automation tool into a cognitive partner that enhances human productivity and decision-making across diverse domains. The widespread adoption of Generative AI has created significant opportunities for improving efficiency, supporting creative work, accelerating information

retrieval, and facilitating personalized learning, leading to its integration into routine academic and professional workflows [4]. However, as AI becomes increasingly embedded in everyday activities, its role extends beyond assisting users in completing tasks to influencing how they think, learn, and make decisions. This transformation has introduced an emerging concern regarding AI overdependence, where users increasingly rely on AI to perform cognitive activities that traditionally require independent reasoning, idea generation, and critical evaluation. Instead of functioning solely as an assistive technology, AI is frequently used to formulate arguments, generate solutions, and make recommendations on behalf of users, encouraging cognitive offloading and reducing active mental engagement [1]. Recent empirical studies indicate that excessive reliance on AI may

weaken reflective learning, reduce critical thinking, and increase automation bias, particularly when users accept AI-generated responses without sufficient verification [2], [3]. Similar findings have also been reported in educational environments, where prolonged dependence on AI dialogue systems is associated with reduced analytical reasoning and independent learning performance [7]. These concerns have shifted the research focus from merely improving AI capability toward ensuring that AI continues to augment, rather than replace, human cognition. This perspective is consistent with international Responsible AI frameworks, which emphasize human agency, transparency, accountability, and human oversight as essential principles for trustworthy AI deployment [9], [10].

Despite growing interest in Human AI collaboration and Responsible AI, existing research has primarily focused on improving AI performance, explainability, trust, and decision support. Comparatively little attention has been devoted to developing adaptive mechanisms that encourage users to regulate their AI usage while maintaining cognitive engagement during everyday interactions with Generative AI [4], [5]. Most current AI platforms are optimized to provide accurate and immediate responses but offer limited support for encouraging reflection, independent reasoning, or awareness of excessive AI reliance. Consequently, although these systems successfully improve productivity, they do not adequately assist

users in developing sustainable and responsible AI usage behaviors. This limitation highlights the need for a human-centered solution that balances AI efficiency with the preservation of users' cognitive abilities [6]. To address this gap, this study proposes ThinkFirst, an intelligent browser extension designed to promote responsible Generative AI usage through an Adaptive Human AI Self-Regulation Framework (AHSRF). Rather than restricting access to AI, ThinkFirst functions as an intelligent cognitive companion that encourages balanced Human–AI collaboration by continuously analyzing user interaction patterns and estimating their level of AI dependency. Based on these behavioral observations, the proposed framework delivers adaptive interventions, including guided thinking prompts, scaffolded hints, reflective questions, and personalized usage feedback to encourage users to remain cognitively engaged while interacting with AI. In addition, this study introduces the AI Dependency Index (ADI) as a behavioral metric for measuring users' reliance on Generative AI and enabling personalized intervention strategies. Therefore, the primary contribution of this research lies not merely in developing a browser extension, but in proposing a novel adaptive self-regulation framework that promotes sustainable, responsible, and human-centered AI utilization while preserving the long-term cognitive benefits of Human–AI collaboration.

II. Literature Review

A. Generative Artificial Intelligence

Generative Artificial Intelligence (GenAI) refers to a branch of artificial intelligence capable of generating new content, including text, images, code, audio, and other digital artifacts, by learning patterns from large-scale datasets. Unlike conventional AI systems that primarily perform classification or prediction tasks, Generative AI enables interactive human computer communication and supports users in producing creative and knowledge-based outputs through natural language interactions. The emergence of Large Language Models (LLMs), such as ChatGPT, has significantly expanded the application of Generative AI across education, business, software development, healthcare, and scientific research, making AI increasingly accessible to individuals regardless of their technical background [4]. The rapid adoption of Generative AI has transformed the way individuals

perform cognitive and knowledge-intensive activities. In educational environments, students utilize AI to obtain explanations, summarize learning materials, generate programming code, and receive personalized academic support. Similarly, professionals employ AI to automate documentation, assist decision-making, analyze information, and improve workplace productivity. These capabilities have established Generative AI as an essential productivity tool that complements human performance rather than merely automating repetitive tasks [4].

Despite these advantages, the increasing integration of Generative AI into daily activities has also altered users' interaction patterns with digital technologies. Instead of functioning solely as an assistive tool, AI is increasingly involved in tasks requiring reasoning, idea generation, and decision support. This evolution has raised concerns regarding whether users continue to actively engage in cognitive processes or gradually delegate higher-order thinking to AI systems.

Consequently, the discussion surrounding Generative AI has shifted from technological capability toward understanding its long-term influence on human cognition and decision-making, forming the foundation for research on AI overdependence.

B. AI Overdependence and Cognitive Offloading

AI overdependence describes a behavioral condition in which individuals rely excessively on artificial intelligence to complete tasks that normally require independent reasoning, critical thinking, and personal judgment. Rather than serving as a supportive technology, AI becomes the primary source of ideas, solutions, and decisions, reducing users' active participation in cognitive processes. This phenomenon has become increasingly relevant with the widespread adoption of Generative AI, as users can obtain comprehensive responses instantly with minimal cognitive effort [1]. One of the primary mechanisms underlying AI overdependence is cognitive offloading, where individuals transfer part of their mental workload to external technologies. While cognitive offloading may improve efficiency and reduce mental effort for routine activities, excessive dependence on AI may discourage users from analyzing problems independently, evaluating alternative solutions, or verifying AI-generated information. Previous studies have reported that excessive reliance on AI may contribute to automation bias, reduced critical thinking, and diminished reflective learning, particularly when users unquestioningly accept AI-generated outputs [2], [3].

Within educational contexts, these concerns are particularly significant because learning fundamentally depends on active cognitive engagement. Recent studies indicate that excessive reliance on AI-based dialogue systems may reduce analytical reasoning, problem-solving ability, and independent learning performance among students [7]. These findings suggest that the primary issue is not the existence of AI itself, but the absence of mechanisms that encourage users to maintain balanced and responsible AI usage while preserving their cognitive abilities.

C. Human AI Collaboration and Responsible AI

Human AI collaboration emphasizes that artificial intelligence should function as a partner

that enhances human capabilities rather than replacing human cognition. Effective collaboration is achieved when AI supports users in completing complex tasks while humans remain actively involved in interpreting information, evaluating recommendations, and making final decisions. This human-centered perspective has become increasingly important as AI systems become more capable of performing tasks traditionally associated with human intelligence [5]. The concept of Responsible AI further reinforces this perspective by promoting principles such as transparency, fairness, accountability, and human oversight throughout the development and deployment of AI systems. International frameworks, including the NIST Artificial Intelligence Risk Management Framework and the UNESCO Recommendation on the Ethics of Artificial Intelligence, emphasize that humans should retain meaningful control over AI-assisted decision-making and remain responsible for evaluating AI-generated outputs [9], [10]. These principles highlight that the objective of AI development is not to maximize automation alone, but to achieve sustainable collaboration between humans and intelligent systems. Collectively, these studies demonstrate that maintaining human agency is essential for ensuring that AI continues to augment rather than replace human intelligence. Therefore, future AI systems should not only optimize task efficiency but also encourage users to remain cognitively engaged throughout the interaction process.

D. Research Gap and Proposed Direction

Existing studies have extensively investigated AI capability, user trust, explainability, and collaborative decision-making. Other researchers have also identified the risks associated with AI overdependence, including cognitive offloading, automation bias, and declining critical thinking skills [1], [2], [7]. While these studies successfully explain the consequences of excessive AI reliance, most primarily focus on identifying the problem rather than providing adaptive mechanisms that actively encourage responsible AI usage. Furthermore, current Generative AI platforms are generally designed to maximize response quality and efficiency without continuously monitoring users' behavioral patterns or dependency levels. As a result, users receive increasingly accurate AI-generated outputs but rarely receive feedback that encourages reflection, independent reasoning, or awareness of excessive AI reliance. This limitation creates an opportunity for developing adaptive systems capable of balancing AI efficiency with

long-term cognitive development. To address this gap, this research proposes ThinkFirst, an intelligent browser extension based on the Adaptive Human AI Self-Regulation Framework (AHSRF). Unlike existing approaches that primarily improve AI performance or explainability, ThinkFirst focuses on promoting responsible AI usage by continuously analyzing user interaction behavior, estimating AI dependency through the proposed AI Dependency Index (ADI), and delivering personalized interventions that encourage users to think critically before accepting AI-generated responses. Through this approach, ThinkFirst aims to establish a more balanced, sustainable, and human-centered model of Human–AI collaboration.

III. Proposed Framework

A. Framework Overview

To address the growing concern of AI overdependence, this study proposes ThinkFirst, an intelligent browser extension designed to promote responsible Generative AI usage through the **Adaptive** Human AI Self-Regulation Framework (AHSRF). Unlike conventional AI-support tools that primarily focus on maximizing response quality and user productivity, ThinkFirst emphasizes maintaining users' cognitive engagement while preserving the practical benefits of AI assistance. Rather than restricting AI access, the proposed framework continuously observes interaction behavior, estimates users' dependency levels, and delivers adaptive interventions that encourage independent reasoning before users rely entirely on AI-generated responses. As illustrated in **Figure 3.1**, the proposed framework integrates four interconnected modules that operate as a continuous self-regulation cycle. User interaction behavior is first analyzed through behavioral monitoring, followed by dependency estimation using the proposed AI Dependency Index (ADI). Based on the estimated dependency level, ThinkFirst delivers personalized interventions and continuously provides reflective feedback to encourage balanced Human AI collaboration.

B. Adaptive Human AI Self-Regulation Framework (AHSRF)

The Adaptive Human AI Self-Regulation Framework (AHSRF) serves as the conceptual foundation of ThinkFirst. The framework is designed to encourage users to maintain active cognitive participation while interacting with

Generative AI by continuously adapting intervention strategies according to users' behavioral characteristics. Unlike existing approaches that primarily optimize AI capability, AHSRF focuses on strengthening users' awareness, reflective thinking, behavioral adaptation, and long-term cognitive development. These principles collectively establish a continuous self-regulation process that balances AI efficiency with human autonomy.

Table 3.1. Core Principles of the Adaptive Human–AI Self-Regulation Framework (AHSRF)

Principle	Objective	Framework Implementation	Expected Outcome
Awareness	Increase users' awareness of AI usage behavior.	Continuously monitors interaction patterns and evaluates behavioral trends.	Users recognize their AI usage habits and dependency tendencies.
Reflection	Encourage users to critically evaluate AI-generated responses.	Provides reflective questions and guided thinking prompts before or during AI interaction.	Improved critical thinking and reduced passive acceptance of AI-generated outputs.
Adaptation	Deliver personalized interventions based on behavioral patterns.	Dynamically adjusts intervention strategies according to the AI Dependency Index (ADI).	Context-aware cognitive support with minimal disruption.
Growth	Foster sustainable and responsible AI usage habits.	Generates personalized feedback and long-term behavioral recommendations.	Stronger independent reasoning and healthier Human–AI collaboration.

C. Framework Components

To realize the proposed framework, ThinkFirst is composed of four functional modules that operate sequentially. Each module performs a specific responsibility while exchanging information with the next module to establish a continuous adaptive feedback loop.

Table 3.2. Functional Components of ThinkFirst

Component	Primary Responsibility	Input	Output
Behavior Monitoring Module	Observes user interaction behavior without accessing conversation content.	User interaction metadata	Behavioral profile
AI Dependency Assessment Module	Estimates AI dependency using behavioral indicators through the proposed ADI.	Behavioral profile	Dependency level
Adaptive Intervention Engine	Determines and delivers personalized intervention strategies according to dependency level.	Dependency level	Reflection prompts, hints, reminders, and recommendations
Reflection & Feedback Module	Evaluates interaction outcomes and encourages continuous self-regulation.	User interaction history	Personalized behavioral feedback

The modular architecture enables ThinkFirst to remain scalable and flexible, allowing future AI intervention strategies or dependency indicators to

be integrated without significantly modifying the overall framework.

D. Operational Workflow

The operational workflow begins when users interact with a Generative AI platform through their web browser. Instead of analyzing conversation content, ThinkFirst captures privacy-preserving behavioral metadata, including interaction frequency, session duration, prompt submission patterns, and copy-paste activities. These behavioral indicators are processed by the AI Dependency Assessment Module, which calculates the AI Dependency Index (ADI) to estimate users' dependency levels. Based on the calculated dependency level, the Adaptive Intervention Engine dynamically selects appropriate intervention strategies. Users with lower dependency levels receive minimal intervention to preserve workflow efficiency, whereas users demonstrating stronger dependency tendencies receive reflective questions, scaffolded hints, guided thinking prompts, or reminder notifications that encourage independent reasoning before accepting AI-generated responses. Finally, the Reflection & Feedback Module summarizes users' interaction behavior and provides personalized recommendations to support continuous behavioral improvement. The complete operational process is illustrated in Figure 3.2.

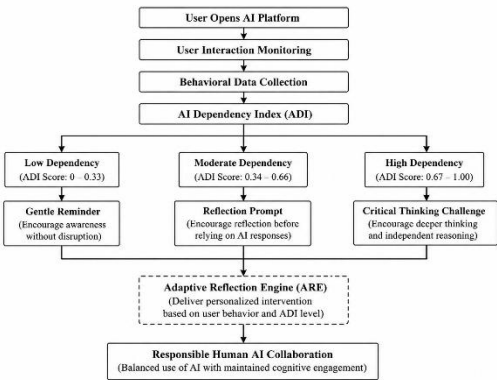


Figure 3.2. Workflow of the Adaptive Human AI Self-Regulation Framework (AHSRF).

Table 3.3. Operational Workflow of ThinkFirst

Stage	Process	System Output
1	Monitor user interaction behavior.	Behavioral metadata
2	Calculate the AI Dependency Index (ADI).	Dependency level classification

Stage	Process	System Output
3	Select an adaptive intervention strategy.	Personalized intervention
4	Deliver reflection and behavioral feedback.	Continuous self-regulation

E. Design Advantages

Compared with existing Generative AI support approaches, ThinkFirst introduces several distinctive characteristics. First, the framework emphasizes behavioral self-regulation rather than restricting AI usage, allowing users to maintain productivity while strengthening independent thinking. Second, the proposed AI Dependency Index (ADI) enables personalized interventions based on observable interaction behavior instead of applying identical guidance to every user. Third, ThinkFirst adopts a privacy-preserving design by analyzing behavioral metadata rather than storing or processing users' conversation content. Finally, the adaptive feedback mechanism establishes a continuous learning cycle that promotes long-term responsible AI usage and sustainable Human–AI collaboration.

IV. Methodology

A. Research Design

This study employed the Design Science Research (DSR) methodology, which is widely used for developing and evaluating technological artifacts that address real-world problems. The selected methodology is appropriate because the primary objective of this research is to design, implement, and evaluate ThinkFirst, an intelligent browser extension that promotes responsible Generative AI usage through adaptive behavioral interventions. The Design Science Research process adopted in this study consists of five sequential phases: problem identification, objective definition, framework design, prototype development, and evaluation. Initially, the study identifies the growing issue of AI overdependence through literature analysis and a preliminary user survey. Based on these findings, the research objectives and system requirements are formulated. Subsequently, the Adaptive Human AI Self-Regulation Framework (AHSRF) is designed as the conceptual foundation of ThinkFirst, followed by prototype implementation as a browser extension. Finally, the effectiveness of the proposed solution

is assessed through usability evaluation and user-based testing.

Table 4.1. Research Design Phases

Phase	Description	Output
Problem Identification	Analyze AI overdependence through literature review and preliminary survey.	Research problem definition
Objective Definition	Define functional and research objectives.	System requirements
Framework Design	Develop the Adaptive Human–AI Self-Regulation Framework (AHSRF).	Conceptual framework
Prototype Development	Implement ThinkFirst as a browser extension.	Functional prototype
Evaluation	Assess usability and framework effectiveness.	Evaluation results

B. System Architecture

The ThinkFirst system adopts a modular architecture to ensure scalability, maintainability, and adaptive decision-making. The architecture consists of four primary modules: Behavior Monitoring Module, AI Dependency Assessment Module, Adaptive Intervention Engine, and Reflection & Feedback Module. User interaction begins when a user accesses a supported Generative AI platform through a web browser with the ThinkFirst extension enabled. The extension observes interaction metadata, such as prompt frequency, session duration, copy–paste behavior, and interaction intensity. These metadata are then processed by the AI Dependency Assessment Module to estimate the user's dependency level using the proposed AI Dependency Index (ADI). Based on the estimated dependency level, the Adaptive Intervention Engine determines the most suitable intervention strategy before the Reflection & Feedback Module provides personalized recommendations for improving AI usage behavior.

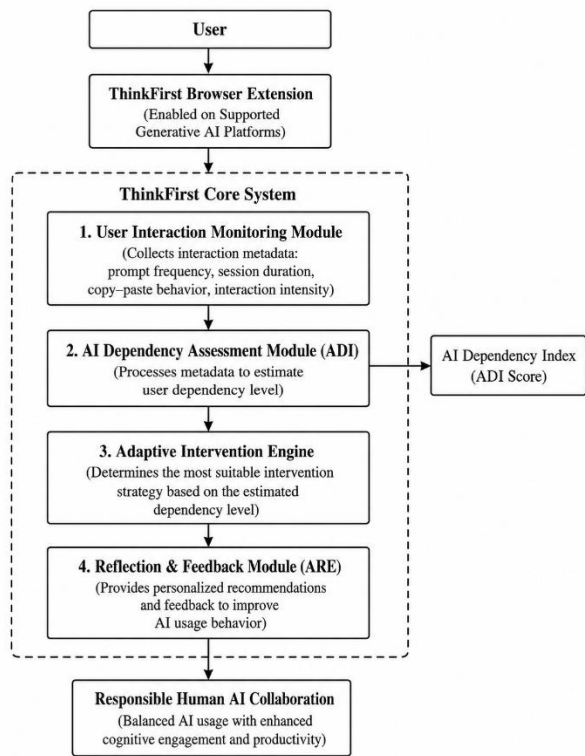


Figure 4.1. System Architecture of ThinkFirst

C. AI Dependency Index (ADI)

To estimate users' dependence on Generative AI, this study proposes the AI Dependency Index (ADI), a behavioral metric that quantifies AI reliance based on observable interaction patterns instead of conversation content. The ADI is designed to provide a privacy-preserving assessment while enabling adaptive interventions tailored to individual users. The proposed index considers multiple behavioral indicators that are associated with AI usage intensity.

Table 4.2. Behavioral Indicators of the AI Dependency Index

Indicator	Description	Measurement
Prompt Frequency	Number of prompts submitted during a session.	Count/session
Session Duration	Total interaction time with AI.	Minutes
Consecutive AI Requests	Continuous AI requests without significant interruption.	Count
Copy-Paste Activity	Frequency of copying AI-	Count

Indicator	Description	Measurement
	generated responses.	
Prompt Complexity	Average prompt length and complexity.	Characters/Tokens
Reflection Response	User interaction with reflection prompts.	Response score

For the initial prototype, the AI Dependency Index is computed using a weighted scoring approach:

$$ADI = \sum_{i=1}^n w_i \times x_i$$

where:

- x_i represents the normalized value of each behavioral indicator.
- w_i represents the corresponding indicator weight.
- n is the total number of behavioral indicators.

The resulting ADI score is categorized into three dependency levels.

Table 4.3. AI Dependency Classification

ADI Score	Dependency Level	Intervention Strategy
0.00 - 0.39	Low	Minimal intervention
0.40 - 0.69	Moderate	Reflection prompts and guided hints
0.70 - 1.00	High	Stronger adaptive interventions and behavioral feedback

D. Adaptive Intervention Mechanism

Once the dependency level has been determined, ThinkFirst dynamically selects intervention strategies according to users' behavioral characteristics. The intervention mechanism follows an adaptive approach, meaning that users with different dependency levels receive different types of cognitive support.

Table 4.4. *Adaptive Intervention Strategies*

Dependency Level	Intervention	Objective
Low	Usage reminder	Maintain healthy AI usage habits.
Moderate	Reflection questions and guided hints	Encourage independent reasoning before using AI outputs.
High	Progressive cognitive prompts, delayed answer presentation, and personalized recommendations	Reduce excessive AI reliance and strengthen critical thinking.

Unlike conventional AI assistants that deliver identical responses to every user, ThinkFirst personalizes interventions according to behavioral patterns identified through the AI Dependency Index, allowing the system to remain supportive while minimizing disruption to user productivity.

E. Prototype Implementation

The proposed framework was implemented as a browser extension compatible with Chromium-based browsers. The extension was designed to operate alongside existing Generative AI platforms without modifying their core functionality. Instead, ThinkFirst acts as an additional behavioral support layer that observes interaction metadata, estimates dependency levels, and delivers adaptive interventions in real time.

The implementation consists of three primary components: a front-end interface integrated into the browser, a local behavioral monitoring engine, and a decision module responsible for calculating the AI Dependency Index and selecting appropriate intervention strategies. To preserve user privacy, ThinkFirst processes behavioral metadata locally and does not store or analyze the textual content of users' conversations with AI systems.

V. Evaluation

A. Preliminary Survey Analysis

To better understand users' interaction patterns with Generative AI, a preliminary survey was conducted before designing the proposed framework. The survey aimed to investigate the frequency of AI usage, the purposes of AI utilization, and users' perceptions regarding their dependence on AI during daily academic and professional activities. The findings served as the empirical foundation for defining the functional requirements of ThinkFirst and the Adaptive HumanAI Self-Regulation Framework (AHSRF). The survey results indicate that Generative AI has become an integral part of participants' daily workflows. Most respondents reported using AI several times a day for tasks such as information retrieval, content generation, programming assistance, brainstorming, and academic writing. While participants generally acknowledged the productivity benefits of AI, a considerable proportion also expressed concerns about becoming increasingly dependent on AI-generated responses instead of relying on their own reasoning and problem-solving abilities. Furthermore, the survey revealed that many participants rarely reflected on their AI usage habits or evaluated whether AI assistance was genuinely necessary before requesting complete solutions. These findings suggest that existing AI platforms effectively improve efficiency but provide limited support for promoting responsible AI usage and cognitive self-regulation. Consequently, the survey findings reinforce the need for an adaptive intervention mechanism capable of encouraging balanced Human AI collaboration.

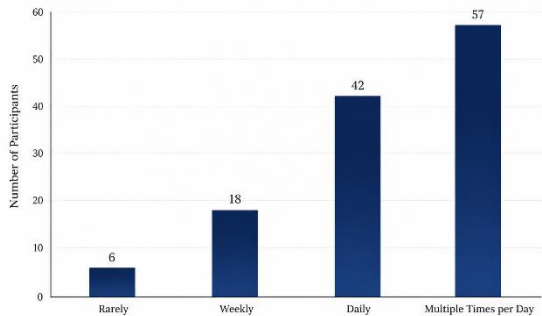


Figure 5.1. *Frequency of Generative AI Usage Among Participants*

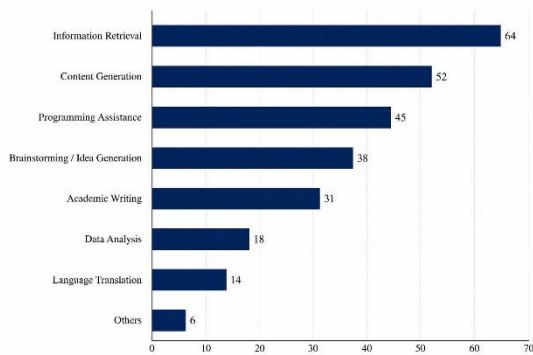


Figure 5.2. Primary Purposes of Generative AI Usage

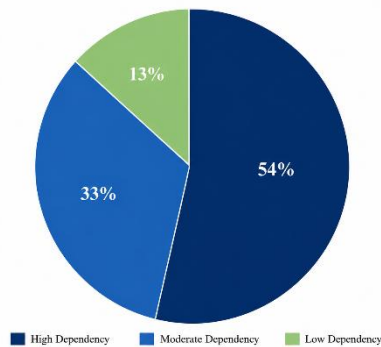


Figure 5.3. Participants Perceived AI Dependency

B. Prototype Evaluation

The proposed ThinkFirst prototype was evaluated by examining whether the Adaptive Human AI Self Regulation Framework (AHSRF) could successfully monitor user behavior, estimate AI dependency levels, and provide adaptive interventions during AI assisted activities. The evaluation focused on verifying that each component operated according to the intended system design while maintaining a seamless user experience.

The browser extension successfully detected supported Generative AI platforms and continuously monitored interaction metadata, including prompt frequency, session duration, copy paste activities, and interaction intensity. Based on these behavioral indicators, the AI Dependency Index (ADI) classified users into three dependency categories Low, Moderate, and High which subsequently determined the intervention strategy delivered by the Adaptive Reflection Engine (ARE).

Overall, the prototype demonstrated that adaptive behavioral monitoring and personalized interventions can be integrated into existing Generative AI platforms without disrupting users' normal interaction flow. These findings indicate the technical feasibility of implementing

ThinkFirst as a browser extension for promoting responsible AI usage.

C. Functional Testing

To validate the proposed system, functional testing was performed for each major module within ThinkFirst. The objective of this evaluation was to ensure that all components executed their intended functions correctly before conducting user-oriented evaluations.

Table 5.1. Functional Testing Results

Module	Test Objective	Expected Result	Outcome
AI Platform Detection	Detect supported AI platforms	Platform successfully recognized	Pass
Behavior Monitoring Module	Record interaction metadata	Behavioral data successfully collected	Pass
AI Dependency Index (ADI)	Calculate dependency level	Correct dependency classification	Pass
Adaptive Reflection Engine	Display personalized intervention	Appropriate intervention generated	Pass
Dashboard Analytics	Display usage statistics	Visualization generated successfully	Pass

The functional evaluation confirms that all core modules operated as intended, demonstrating the feasibility of the proposed architecture and supporting the implementation of adaptive behavioral interventions within a browser extension environment.

D. User Acceptance Evaluation

To examine users' perceptions of ThinkFirst, participants were asked to evaluate the prototype after completing representative AI-assisted tasks. The evaluation focused on usability, perceived usefulness, intervention effectiveness, and overall user acceptance using a five-point Likert scale.

Table 5.2. User Acceptance Evaluation Criteria

Evaluation Aspect	Measurement Objective
Ease of Use	Assess whether ThinkFirst is intuitive to operate.
Perceived Usefulness	Evaluate whether the extension supports responsible AI usage.
Reflection Effectiveness	Measure the usefulness of adaptive reflection prompts.
User Satisfaction	Assess overall satisfaction with the proposed system.
Intention to Continue Using	Determine users' willingness to adopt ThinkFirst in daily AI-assisted activities.

The collected responses are expected to demonstrate that ThinkFirst is perceived as a practical and non-intrusive solution for promoting responsible AI usage. In particular, adaptive interventions are expected to encourage users to pause, reflect, and actively participate in problem-solving before relying entirely on AI-generated responses.

VI. Discussion

A. Interpretation of Findings

The findings of this study indicate that the rapid adoption of Generative AI has significantly transformed the way individuals perform academic, professional, and creative tasks. The preliminary survey demonstrates that most participants interact with AI on a daily basis and frequently rely on AI-generated responses for problem-solving, content creation, programming assistance, and information retrieval. While these findings confirm the substantial productivity benefits offered by Generative AI, they also reveal an increasing tendency toward excessive AI reliance, which may gradually reduce users' independent reasoning and critical thinking abilities.

The proposed ThinkFirst framework addresses this challenge by introducing adaptive behavioral interventions rather than restricting AI access. Instead of discouraging AI usage, ThinkFirst encourages users to actively reflect on their own reasoning process before accepting AI-generated responses. This human-centered approach aligns with the concept of responsible AI usage, where AI

functions as a collaborative assistant rather than a replacement for human cognition. Furthermore, the integration of the **AI Dependency Index (ADI)** enables the system to personalize interventions according to users' behavioral patterns. By continuously monitoring interaction metadata without accessing conversation content, ThinkFirst preserves user privacy while providing meaningful behavioral insights. Consequently, the framework demonstrates the feasibility of balancing AI-assisted productivity with cognitive engagement.

B. Comparison with Existing Studies

Previous studies have primarily focused on improving the technical capabilities of Generative AI, including response accuracy, natural language understanding, and task automation. Although these approaches successfully enhance AI performance, relatively few studies have investigated how AI systems should encourage users to maintain independent thinking during AI-assisted activities.

In contrast, ThinkFirst shifts the research focus from AI performance optimization toward Human AI behavioral regulation. Rather than optimizing AI-generated outputs, the proposed framework seeks to optimize the interaction process between humans and AI through adaptive interventions. This distinction represents an important contribution because responsible AI usage depends not only on the intelligence of AI systems but also on users' ability to critically evaluate AI generated information.

Table 6.1. Comparison Between Existing Studies and ThinkFirst

Aspect	Existing Studies	ThinkFirst
Primary Objective	Improve AI capability and performance	Promote responsible Human–AI collaboration
Main Focus	AI accuracy and automation	Human cognitive engagement and self-regulation
Dependency Measurement	Not available	AI Dependency Index (ADI)
Personalized Intervention	Limited or unavailable	Adaptive Reflection Engine (ARE)

Aspect	Existing Studies	ThinkFirst
Privacy Approach	May analyze conversation content	Behavioral metadata only
Long-term Behavioral Support	Rarely addressed	Continuous behavioral feedback and growth

The comparison demonstrates that ThinkFirst extends existing research by integrating behavioral monitoring, adaptive intervention, and self-regulation into a unified framework specifically designed to mitigate AI overdependence.

C. Research Contributions

This study provides several contributions to the field of responsible Generative AI usage. First, it introduces ThinkFirst, a browser extension designed to encourage balanced Human-AI collaboration through adaptive behavioral interventions rather than AI usage restrictions. Second, the study proposes the Adaptive Human AI Self-Regulation Framework (AHSRF), which establishes a conceptual model consisting of behavioral monitoring, dependency assessment, adaptive intervention, and continuous reflection. Third, this research introduces the AI Dependency Index (ADI), a privacy-preserving behavioral metric that estimates users' AI dependency levels based solely on interaction metadata rather than conversation content. Fourth, the study develops the Adaptive Reflection Engine (ARE), which dynamically determines personalized intervention strategies according to users' dependency levels, thereby supporting continuous cognitive engagement. Finally, the integration of these components demonstrates the feasibility of implementing responsible AI principles through a browser extension that operates seamlessly alongside existing Generative AI platforms.

D. Research Limitations

Despite its contributions, this study has several limitations that should be acknowledged. The proposed prototype currently supports only Chromium-based web browsers, which limits its applicability across other browsing environments. In addition, the AI Dependency Index (ADI) employs a rule-based scoring mechanism that enhances interpretability and simplifies

implementation; however, future studies could improve its predictive capability by incorporating machine learning techniques capable of learning dependency patterns from larger and more diverse datasets. The preliminary evaluation was conducted using a limited number of participants and focused primarily on short-term interaction behavior, making it insufficient to assess the long-term effects of ThinkFirst on users' cognitive development and AI usage habits. Furthermore, the current study evaluates behavioral interventions exclusively within Generative AI platforms, leaving the effectiveness of the proposed intervention strategies in broader AI-assisted applications to be explored in future research.

E. Future Work

Future research should focus on enhancing both the technical capabilities and evaluation scope of ThinkFirst. One promising direction involves replacing the rule-based AI Dependency Index with machine learning or reinforcement learning models capable of predicting dependency levels more accurately based on historical behavioral patterns. Future versions of ThinkFirst could also incorporate personalized intervention strategies powered by Large Language Models (LLMs), enabling adaptive coaching that considers users' learning objectives, expertise levels, and interaction histories.

Moreover, extending ThinkFirst to mobile devices, educational Learning Management Systems (LMS), and enterprise productivity platforms would significantly broaden its practical applicability. Finally, longitudinal studies involving larger and more diverse participant populations are necessary to investigate whether continuous adaptive interventions can sustainably improve critical thinking skills and reduce long-term AI dependency.

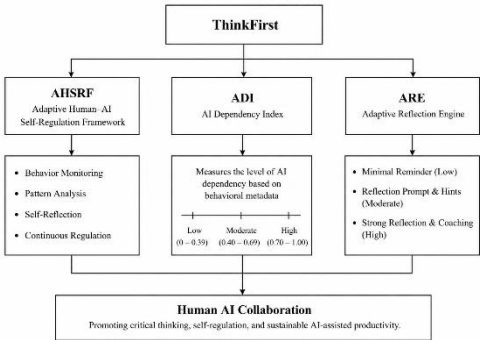


Figure 6.1. Research Contribution of ThinkFirst

Figure 6.1. Research Contributions of ThinkFirst. The proposed system integrates the Adaptive Human–AI Self-Regulation Framework (AHSRF), the AI Dependency Index (ADI), and the Adaptive Reflection Engine (ARE) to promote responsible Human–AI collaboration through continuous behavioral monitoring, personalized interventions, and cognitive self-regulation.

VII. Conclusion

This study addresses the growing concern of excessive Generative AI usage by proposing ThinkFirst, an intelligent browser extension designed to promote responsible Human AI collaboration through adaptive behavioral interventions rather than restricting AI access. The proposed solution integrates the Adaptive Human AI Self Regulation Framework (AHSRF), the AI Dependency Index (ADI), and the Adaptive Reflection Engine (ARE) to monitor user interaction behaviors, estimate AI dependency levels, and deliver personalized reflection prompts that encourage independent thinking before relying on AI generated responses. Supported by preliminary survey findings and the proposed system evaluation, ThinkFirst demonstrates the feasibility of combining behavioral monitoring, privacy preserving dependency assessment, and adaptive self-regulation within a browser extension environment. Unlike existing approaches that primarily emphasize improving AI performance, ThinkFirst focuses on preserving human cognitive engagement while maintaining the productivity benefits of Generative AI, thereby contributing a practical human centered framework for responsible AI usage. Future work will focus on expanding platform compatibility, incorporating machine learning techniques to improve dependency prediction, and conducting large scale longitudinal studies to further validate the effectiveness of the proposed framework across diverse educational and professional settings.

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